University of Chittagong

Department of Computer Science and Engineering

7^{th} Semester B.Sc. (Engineering) Examination 2024

Course Title: Computer Graphics

Course Code: CSE 715

Full Marks: 54 Duration: 4:00 Hours

Answer any **three** questions from each section.

[All parts of a question should be answered sequentially; figures in the right hand margin indicate full marks]

Section A

1. a) What are the key components and technologies involved in transforming simple geometric shapes into complex, realistic computer-generated scenes? 2

   b) What is geometric representation in computer graphics, and why is it important? 2

   c) What is hidden surface removal in computer graphics, and what problem does it solve during image generation? 3

   d) Differentiate between object space and image space in computer graphics, and explain how each is used in the process of generating an image. 2

2. a) Explain the concept of a color model in computer graphics and why different models, such as RGB and CMY, are used for image representation. 3

   b) If a 128 × 128 region is to be cropped from the center of a 512 × 512 image, what are the pixel coordinates of its upper-left corner in the original image? 2

   c) An image uses 10-bit grayscale pixel values in a lookup table. How many entries are required in the lookup table? 1

   d) What is halftoning in computer graphics, and how does it create the appearance of continuous-tone images using a limited number of shades? 3

3. a)

   Examine the halftone pattern in the above image. Describe how the dot size and spacing change across the gradient and explain how this creates the perception of varying intensity levels? 2

   b) Given 2x2 dither matrix

$$D_2 = \begin{bmatrix} 0 & 2 \\ 3 & 1 \end{bmatrix}$$

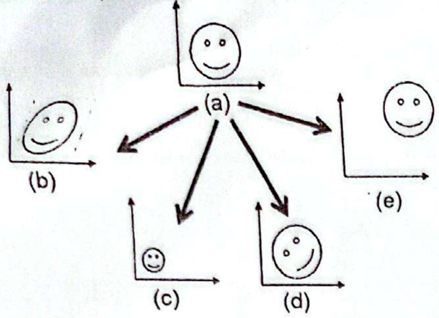
   Use the recurrence relation

$$D_{2n} = \begin{bmatrix} 4D_n & 4D_n + 2 \\ 4D_n + 3 & 4D_n + 1 \end{bmatrix}$$

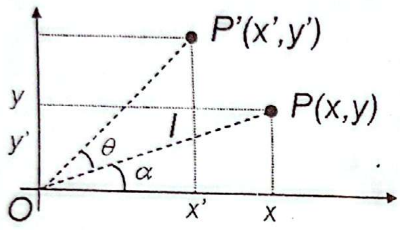
   Construct 4x4 dither matrix D_4. 3

   c) A line is to be drawn from (2,3) to (10,7).
      1. Predict the next pixel using the DDA algorithm if the current pixel is (2,3). Show your calculation for the increments Δx and Δy.
      2. Predict the next pixel using Bresenham's algorithm if the current pixel is (2,3). Compute the initial decision parameter and determine the next pixel.
      3. Continue step by step until the line reaches (10,7), recording the pixel coordinates for both algorithms. 4
      4. At each step, checks are DDA and Bresenham choosing the same pixel? If not, explain why.

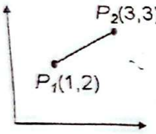
4. a) Given a shape in a 2D coordinate system in Figure (a), mention the transformations used on (a) to generate the shapes in (b), (c), (d) and (e). 2



b) A point P(x,y) is rotated at an angle θ about the origin O in count-clockwise direction. Derive the rotation matrix using the following figure. 2



c) Translate the given line segment $P_1 P_2$ by -1 units in the x direction and -2 units in the y direction. What are the coordinates of the end points after translation? 2



d) Mentions the steps needed to rotate an arbitrary line about a point P_1. 2

e) Write down the matrices for shear transformation in a 2D space. 1

Section B

5. a) Find the normalization transformation that maps a window with lower-left corner at (1, 1) and upper-right corner at (3, 5) onto: 3
(i) A viewport that is the entire normalized device screen (i.e., from (0, 0) to (1, 1)).
(ii) A viewport that has lower-left corner at (0, 0) and upper-right corner at (v_x, v_y).

($\frac{1}{2}$, $\frac{1}{2}$)

b) A clipping window is defined by: 3
$x_{min} = 10, x_{max} = 30$, $y_{min} = 10$, $y_{max} = 25$
For each of the following points, compute the 4-bit Cohen–Sutherland region code (TOP, BOTTOM, RIGHT, LEFT in this exact bit order):
1. P1=(5,5) 2. P2=(20,30) 3. P3=(35,15) 4. P4=(15,8) 5. P5=(12,20)
Then answer: Using the region codes, determine whether the line segment joining A(5,5)and B(12,20) is visible, not visible or requires clipping.

c) A circular clipping window has its center at C (10, 10). 3
Radius r=6, a point p1 (2, 8) another point is P2 (8, 16).
Determine whether the line segment is:
Completely inside the circle, completely outside, or intersecting the circular window. If intersections occur, compute the intersection point(s) and give the clipped visible portion of the line segment.

6. a) Given the following polygons: 3
Clipping window: Convex quadrilateral with vertices A(1,1), B(5,1), C(5,4), D(1,4)
Clipping polygon (convex window): Triangle with vertices P(2,2), Q(4,2), R(3,5)
Perform Weiler-Atherton polygon clipping to determine the resulting clipped polygon.
i) Show entry and exit points.
ii) List the vertices of the clipped polygon in counter-clockwise order.

b) Define tilting as a rotation about the x axis followed by a rotation about the y axis: (i) find the tilting matrix; (ii) does the order of performing the rotation matter? 3

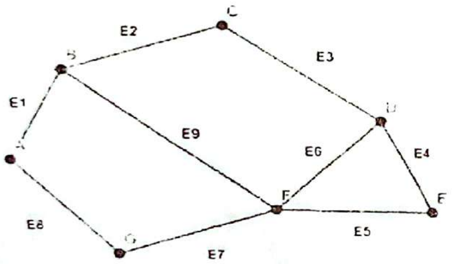
c) What is the significance of Axonometric projection? 2

d) Demonstrate some properties of Cabinet projection. 1

7. a) Define visible surface and back-face with necessary examples. 2

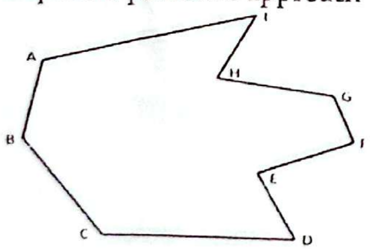
b) Mention the steps of polygon drawing using trapezoid primitive approach. 2

c) Write down the vertex, edge and surface matrices of the given polygon. 2

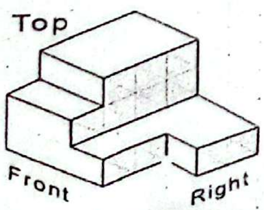


d) Illustrate z-buffer algorithm with necessary figures. What are its advantages and disadvantages? 2

e) How many trapezoids will be formed for the given polygon for polygon filling using trapezoid primitive approach 1



8. a) 3



Draw top, front and right/side view.

b) Why Ray Tracing is efficient method of generating images? 2

c) Distinguish between ray and vector. 2

d) Given the plane $5x - 3y + 6z = 7$: (i) find the normal vector to the plane, and (ii) determine whether P1 (1, 4, 2) and P2 (−5, −1, 3) are on the same side of the plane. 2